



MASTER-KIDS · OLYMPIAD PRACTICE SERIES · CLASS 4

Mathematics Olympiad Worksheet 36

Level 2 — Intermediate · Geometry I

Class	Class 4 (Primary · Stage 2)	Subject	Mathematics
Level	Level 2 — Intermediate	Topic Focus	Geometry I — Angles, Lines, Symmetry & Circles
Questions	20	Time Allowed	35 minutes
Total Marks	20	Worksheet No.	36 of 100
Curriculum	CBSE · ICSE · State Boards (board-agnostic) · NEP 2020-aligned		

About Master-Kids

Master-Kids is India's cradle-to-career learning platform — one platform that grows with a child from Pre-Nursery all the way to college, turning everyday effort into visible progress for parents and a rewarding game of streaks, XP and badges for children. This Olympiad practice series is part of the Master-Kids Primary School programme (Stage 2, Class 1-5). It is board-agnostic — suited to CBSE, ICSE and State Board children alike — and aligned with the NEP 2020 focus on foundational numeracy, science thinking and language.

About This Worksheet

Worksheet 36 brings in the geometry vocabulary the IMO expects at Level 2: clock-hand angles in degrees (30° per hour-gap), parallel vs perpendicular lines, lines of symmetry in shapes AND letters (the rectangle-has-2-not-4 trap), radius and diameter, and triangle types. Symmetry of letters is a recurring SOF favourite.

Learning Objectives

- Measure clock-hand angles using 30° per hour-gap.
- Distinguish parallel and perpendicular lines in real objects.
- Count lines of symmetry in shapes and capital letters.
- Relate radius and diameter ($d = 2r$) both ways.
- Name triangle types and angle sizes including 180° and 360° .

Instructions

1. This worksheet has 20 multiple-choice questions. Each question has exactly one correct answer.
2. Read every question carefully. Look at any picture, table or figure before choosing.
3. Each correct answer earns +1 mark. There is no negative marking, so attempt every question.
4. Write your chosen option — (A), (B), (C) or (D) — clearly in the answer box or on a separate sheet.
5. Do all rough work in the margin or on rough paper. Do not use a calculator.
6. After finishing, check your answers against the Answer Key, then read the Detailed Solutions to understand the reasoning behind every question — that is where the real learning happens.

Marking scheme: +1 mark for each correct answer · No negative marking · Maximum 20 marks.

SECTION A · Multiple Choice Questions · 20 Questions

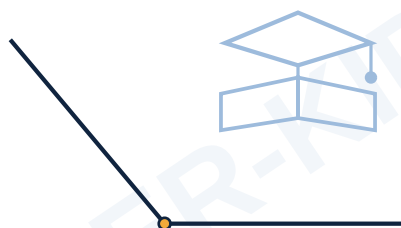
Q1. At **6:00**, the hands of a clock form a straight angle. How many DEGREES is that?

- (A) 90° (B) 120°
 (C) 180° (D) 360°

Q2. The hands of a clock at **4:00** make an angle of:

- (A) 90° (B) 100°
 (C) 120° (D) 150°

Q3. Look at the angle shown. Which type is it?



- (A) Acute (B) Right
 (C) Obtuse (D) Straight

Q4. How many RIGHT angles are there inside a rectangle?

- (A) 2 (B) 3
 (C) 4 (D) 6

Q5. Two lines that NEVER meet, like railway tracks, are called:

- (A) Perpendicular lines (B) Parallel lines
 (C) Curved lines (D) Crossing lines

Q6. Two lines that meet to form a RIGHT angle, like a plus sign (+), are called:

- (A) Parallel lines (B) Perpendicular lines
 (C) Equal lines (D) Slanting lines

Q7. How many lines of symmetry does a RECTANGLE (not a square) have?

- (A) 1 (B) 2
 (C) 4 (D) 0

Q8. How many lines of symmetry does the capital letter **H** have?

- (A) 0 (B) 1
 (C) 2 (D) 4

Q9. Which capital letter has NO line of symmetry?

- (A) A (B) M
 (C) F (D) O



Q10. The distance from the CENTRE of a circle to any point on the circle is called the:

- (A) Diameter (B) Radius
(C) Boundary (D) Chord

Q11. The radius of a circle is 7 cm. Its DIAMETER is:

- (A) 3.5 cm (B) 7 cm
(C) 14 cm (D) 21 cm

Q12. The diameter of a wheel is 50 cm. Its RADIUS is:

- (A) 100 cm (B) 50 cm
(C) 25 cm (D) 20 cm

Q13. A triangle with ALL THREE sides equal is called:

- (A) A right triangle (B) An equilateral triangle
(C) A scalene triangle (D) An isosceles triangle

Q14. How many angles SMALLER than a right angle (acute angles) can you see in the letter **V**?

- (A) 0 (B) 1
(C) 2 (D) 3

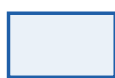
Q15. A square has 4 lines of symmetry. A circle has:

- (A) 4 lines (B) 8 lines
(C) No lines (D) Countless (unlimited) lines

Q16. Which time on a clock makes a RIGHT angle between the hands?

- (A) 6:00 (B) 3:00
(C) 1:00 (D) 12:00

Q17. In which figure are the opposite sides parallel AND all sides equal?



(A)



(B)



(C)



(D)

- (A) Shape A (B) Shape B
(C) Shape C (D) Shape D

Q18. An angle of exactly **360°** means:

- (A) A right angle (B) Half a turn
(C) A complete full turn (D) No turn at all

Q19. How many lines of symmetry does an EQUILATERAL triangle have?

- (A) 1 (B) 2
(C) 3 (D) 0



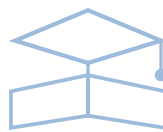
Q20. A ladder leans against a wall. The wall is vertical and the ground is horizontal. The wall and ground are ___ to each other.

(A) Parallel

(B) Perpendicular

(C) Equal

(D) Curved



ANSWER KEY & DETAILED SOLUTIONS

Quick Answer Key

Q	1	2	3	4	5	6	7	8	9	10
Ans	C	C	C	C	B	B	B	C	C	B
Q	11	12	13	14	15	16	17	18	19	20
Ans	C	C	B	B	D	B	B	C	C	B

Step-by-Step Solutions

Q1. Correct answer: (C)

A straight angle (a straight line) measures **180°**.

Q2. Correct answer: (C)

From 12 to 4 is 4 hour-gaps; each gap is 30°: $4 \times 30^\circ = 120^\circ$ (an obtuse angle).

Q3. Correct answer: (C)

The angle opens wider than 90° but less than 180° — an **obtuse** angle.

Q4. Correct answer: (C)

All **4** corners of a rectangle are right angles (90° each).

Q5. Correct answer: (B)

Lines that stay the same distance apart and never meet are **parallel lines**.

Q6. Correct answer: (B)

Lines meeting at 90° are **perpendicular**.

Q7. Correct answer: (B)

A rectangle folds onto itself in **2** ways — through the midpoints of opposite sides. (Its diagonals are NOT lines of symmetry; only a square has 4.)

Q8. Correct answer: (C)

H has a vertical line of symmetry AND a horizontal one — **2 lines**.

Q9. Correct answer: (C)

F cannot be folded to match itself. (A and M have a vertical line; O has many.)

Q10. Correct answer: (B)

Centre to circle = **radius**. (The diameter goes all the way across, through the centre.)

Q11. Correct answer: (C)

Diameter = $2 \times \text{radius} = 2 \times 7 = 14 \text{ cm}$.

Q12. Correct answer: (C)

Radius = diameter $\div 2 = 50 \div 2 = 25 \text{ cm}$.

Q13. Correct answer: (B)

All sides equal = **equilateral** triangle. (Two equal sides = isosceles; all different = scalene.)

Q14. Correct answer: (B)

The letter V forms **1** acute angle at its point.

Q15. Correct answer: (D)

A circle can be folded along ANY line through its centre — it has **countless lines of symmetry**.

**Q16. Correct answer: (B)**

At **3:00** the hands point to 12 and 3 — a quarter turn = 90° . (6:00 gives 180° ; 12:00 gives 0° .)

Q17. Correct answer: (B)

The **square (Shape B)** has opposite sides parallel and all four sides equal. (The rectangle has parallel sides but unequal lengths and widths.)

Q18. Correct answer: (C)

360° is one **complete full turn** — back to where you started. (180° is half a turn.)

Q19. Correct answer: (C)

An equilateral triangle has **3** lines of symmetry — one through each corner to the middle of the opposite side.

Q20. Correct answer: (B)

The wall stands at 90° to the ground — they are **perpendicular**.

